

Series

Example: $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots + \frac{1}{n^4} + \dots = \frac{\pi^4}{90}$

$a_1 \quad a_2 \quad a_3 \quad a_n$

← e.g.

Series basics

- Individual contributions are terms of the series, often presented with a subscript, e.g. $a_n, t_j, \dots$. They form a sequence.
- Finite and Infinite Series: If we have an infinite series $S = a_1 + a_2 + a_3 + \dots = \sum_{n=1}^{\infty} a_n$, then $S_N = \sum_{n=1}^N a_n = a_1 + a_2 + \dots + a_N$ is called a partial sum of S .
- If $S = \sum_{n=1}^{\infty} a_n$, the terms a_n are either given:
 - by explicitly listing their values, e.g. $a_1 = 7, a_2 = 9, a_3 = 11, \dots$
 - by explicit formulas, e.g. $a_k = \left(\frac{1}{16}\right)^k$
 - by recurrence relations, e.g. $a_k = \frac{1}{16} a_{k-1}, a_0 = 1$
- The sum of the series is the value of the series: $\lim_{N \rightarrow \infty} S_N$

Convergence and Divergence

- A finite series has a finite sum
- For an infinite series, we consider its partial sums $S_N = a_1 + a_2 + \dots + a_N$ and $S = \sum_{k=1}^{\infty} a_k$ is convergent if $\lim_{N \rightarrow \infty} S_N$ exists, and is divergent if not convergent.
- A necessary condition for a series $S = \sum_{j=0}^{\infty} b_j = b_0 + b_1 + \dots$ to converge is that $b_j \rightarrow 0$ as $j \rightarrow \infty$

Examples:

Arithmetic Series $S = \sum_{n=1}^{\infty} n$ is divergent. Consider the N th partial sum $S_N = \sum_{n=1}^N n$

$$S_N = 1 + 2 + 3 + \dots + (N-1) + N$$

$$S_N = N + (N-1) + (N-2) + \dots + 2 + 1$$

$$2S_N = N + N + \dots + N + N = N(N+1) \Rightarrow S_N = \frac{N(N+1)}{2}$$

$$T_N = 1^2 + 2^2 + \dots + N^2 = \frac{N(N+1)(2N+1)}{6}$$

Geometric Series

$$a \neq 0, r \in \mathbb{R}$$

$$S = \sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \dots$$

$$\text{Partial Sum } S_N = \sum_{n=0}^N ar^n = a + ar + \dots + ar^N$$

$$r S_N = ar + ar^2 + ar^3 + \dots + ar^N + ar^{N+1}$$

multiply both sides by r

$$\Rightarrow S_N - r S_N = a - ar^{N+1} = a(1 - r^{N+1}) = (1-r)S_N$$

If $r \neq 1$ we can divide by $1-r$

$$S_N = \frac{a(1-r^{N+1})}{1-r}$$

$$\lim_{N \rightarrow \infty} \frac{a(1-r^{N+1})}{1-r} \text{ exists if and only if } |r| < 1$$

Important: the geometric series $S = \sum_{n=0}^{\infty} ar^n$ with $a \neq 0$ converges for all $|r| < 1$ and diverges for all $|r| \geq 1$

$$\text{Consider } \sum_{n=n_0}^{\infty} r^n, n_0 \in \mathbb{N}, |r| < 1$$

$$\text{Using } S = \frac{a(1 - \lim_{N \rightarrow \infty} r^N)}{1-r} = \frac{a}{1-r}$$

$$\begin{aligned} \sum_{n=n_0}^{\infty} r^n &= \sum_{m=0}^{\infty} r^{n_0+m} \\ &= r^{n_0} \sum_{m=0}^{\infty} r^m = r^{n_0} \cdot \frac{1}{1-r} = \frac{r^{n_0}}{1-r} \end{aligned}$$

Harmonic Series

$$S = \sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

$\sum a_k \Rightarrow a_k = \frac{1}{k}$ which goes to 0 as $k \rightarrow \infty$
but does NOT converge

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \frac{1}{10} + \dots + \frac{1}{16}\right) + \dots + \left(\frac{1}{2^n+1} + \frac{1}{2^n+2} + \dots + \frac{1}{2^{n+1}}\right)$$

2 terms $> \frac{1}{4}$, 4 terms $> \frac{1}{8}$, 8 terms $> \frac{1}{16}$, 2^n terms $> \frac{1}{2^n}$

$\therefore$ Sum goes to $+\infty \rightarrow$ divergent.

Note $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$

Finding The Sum

Algebraic Manipulations

$$\sum k a_n = k \sum a_n = S$$

$$\sum_n a_n = S \text{ and } \sum_n b_n = T \text{ then } \sum_n (a_n + b_n) = S + T.$$

Example: $a_n = a + nb$

$$\begin{aligned} S &= \sum_{n=1}^N a_n = \sum_{n=1}^N (a + nb) \\ &= \sum_{n=1}^N a + \sum_{n=1}^N nb \\ &= a \sum_{n=1}^N 1 + Nb \\ &= \frac{aN(N+1)}{2} + Nb \\ &= N \left(\frac{a(N+1)}{2} + b \right) \end{aligned}$$

Method of Differences

$$S = \sum_{k=1}^N a_k \quad (N \text{ finite or infinite})$$

Assume we find b_k such that $a_k = b_{k+1} - b_k$ for all k

$$\begin{aligned} \text{Then we obtain for the partial sums } S_n &= \sum_{k=1}^n a_k = \sum_{k=1}^n (b_{k+1} - b_k) = \sum_{k=1}^n b_{k+1} - \sum_{k=1}^n b_k \\ &= \sum_{k=2}^{n+1} b_k - \sum_{k=1}^n b_k \\ &= b_{n+1} - b_1 \end{aligned}$$

If we have $\lim_{n \rightarrow \infty} b_n = b$, then $\lim_{n \rightarrow \infty} S_n = b - b_1 = \sum_{n=1}^{\infty} a_n$

Generalization: If $a_k = b_{k+m} - b_k$ $m \geq 1, m \in \mathbb{Z}$ then:

$$S_n = b_{n+1} + b_{n+2} + \dots + b_{n+m} - (b_1 + b_2 + \dots + b_m)$$

If $b_n \rightarrow b$ as $n \rightarrow \infty$, then $S = \lim_{n \rightarrow \infty} S_n = m \cdot b - (b_1 + b_2 + \dots + b_m)$

bigger m terms all tend to a value b , just treat them all as equal to b .

Example: $S = \sum_{k=1}^{\infty} \frac{1}{k(k+m)}$

Want $\frac{1}{k(k+m)}$ in the form $b_{k+m} - b_k$

$$\frac{1}{k} - \frac{1}{k+m} = \frac{(k+m) - k}{k(k+m)} = \frac{m}{k(k+m)}$$

$$\therefore \frac{1}{k(k+m)} = -\frac{1}{m} \left(\frac{1}{k+m} - \frac{1}{k} \right)$$

$$\therefore b_k = -\frac{1}{mk}$$

$$= \frac{1}{m(k+m)} + \frac{1}{mk}$$

$$\therefore b_k = \frac{1}{mk}$$

$$b_k = -\frac{1}{mk}$$

$$\lim_{k \rightarrow \infty} b_k = \lim_{k \rightarrow \infty} \left(-\frac{1}{mk} \right) = 0$$

$$\Rightarrow \sum_{k=1}^{\infty} \frac{1}{k(k+m)} = m \cdot 0 + \frac{1}{m} (1 + \frac{1}{2} + \dots + \frac{1}{m})$$

$$= \frac{1}{m} (1 + \frac{1}{2} + \dots + \frac{1}{m})$$

here bigger m terms all tend to 0.

$$\sum_{n=1}^N a_n(n) = S_N(n)$$

Differentiating termwise gives:

$$\sum_{n=1}^{\infty} a'_n(n) = S'_N(n)$$

and we can also integrate termwise

Example : Arithmetic-Geometric Series :

$$\sum_{n=0}^{\infty} n r^n \quad \text{when } n=0, n r^n = 0 \text{ so ignore it!}$$

$$= \sum_{n=1}^{\infty} n r^n$$

Nth partial sum $S_N(r) = \sum_{n=1}^N n r^n = r \sum_{n=1}^N n r^{n-1}$

Can just take the derivative at apparently for funnier

$$= r \sum_{n=1}^N (r^n)' \quad \text{derivative}$$

$$= r \left(\sum_{n=1}^N r^n \right)'$$

$$= r \left(r \sum_{n=0}^{N-1} r^n \right)'$$

$$= r \left(r (1 + r + r^2 + \dots + r^{N-1}) \right)'$$

Using $\sum_{k=0}^M q^k = \frac{1-q^{M+1}}{1-q} \quad (q \neq 1)$

$$S_N(r) = r \left(r \frac{1-r^{N+1}}{1-r} \right)'$$

$r \neq 1$ differentiate

$$= r \left(- (N+1) \frac{r^N}{1-r} + \frac{1-r^{N+1}}{(1-r)^2} \right)$$

$$= r \left(\frac{1-r^N}{(1-r)^2} - \frac{N r^N}{1-r} \right)$$

If $|r| < 1$, we have $\lim_{N \rightarrow \infty} S_N(r) = \frac{r}{1-r^2} = \sum_{n=1}^{\infty} n r^n$

Logarithmic Series : $\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \quad \text{for all } |t| < 1 \quad \left[\text{from } S_n = \frac{a}{1-r} \text{ when } |r| < 1 \right]$

$$\Rightarrow \log(1-t) = - \int_0^t \frac{dt}{1-t}$$

Interchanging $\int$ with infinite sum can be on whole

you can't always do this but it works in this case bc magic

$$= - \int_0^t \sum_{n=0}^{\infty} t^n dt$$

$$= - \sum_{n=0}^{\infty} \int_0^t t^n dt$$

$$= - \sum_{n=0}^{\infty} \left[\frac{t^{n+1}}{n+1} \right]_0^t = - \sum_{n=0}^{\infty} \frac{t^{n+1}}{n+1}$$

$$\stackrel{\text{index shift}}{=} - \sum_{n=1}^{\infty} \frac{1}{n} = - \sum_{n=1}^{\infty} \frac{1}{n} \quad \forall |x| < 1$$

$$\log(1+x) = \log(1-(-x)) = - \sum_{n=1}^{\infty} \frac{(-x)^n}{n} = - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n} \\ = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} \quad \forall |x| < 1$$

Convergence Tests

Often the sums of infinite series are not possible to get explicitly, but may still converge.

e.g. $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

but $\sum_{n=1}^{\infty} \frac{1}{n^3} = ?$

Look only at series $\sum_{n=1}^{\infty} a_n$ with $a_n \geq 0$, i.e. where $S_N = \sum_{n=1}^N a_n \leq S_{N+1}$

Convergence or Divergence of $\sum a_n$ does not change when we change finitely many terms.

1. Comparison Test: Given a series $S = \sum a_n$ find another series $T = \sum b_n$ whose convergence or divergence is known to compare it to. ($a_n, b_n \geq 0 \forall n$)

We assume $a_n \leq b_n$ for all terms for some $n \geq n_0$ onwards.

Then: If T is Convergent, S is Convergent.

If S is Divergent, then T is Divergent too.

Example: $S = \sum_{n=1}^{\infty} \frac{2^n + 5}{3^{n+1}}$ Compare to $T = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ ↗ $b_n \geq 0$

Since $\frac{2^n + 5}{3^{n+1}} \leq \frac{2^n}{3^n}$ for some n big enough this can be confirmed $\rightarrow$ gen. series.

Since $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n - 1 = \frac{1}{1-\frac{2}{3}} - 1 = 3 - 1 = 2 = T$

Changing the 1 to a 0 here adds 1 to the overall sum so this is Subtotal

2. Limit Comparison Test (Equivalent test): Given $S = \sum a_n$ and $T = \sum b_n$
 $(a_n, b_n > 0)$

Consider $\rho = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$:

- If ρ is finite and non-zero, S and T are either both convergent or both divergent.
- If $\rho = 0$, $T > S \rightarrow$ So if T converges, then S converges.
 $\rightarrow$ If S diverges, T must also diverge.
- If ρ does not exist, then we cannot determine anything using this test.

Example: $S = \sum_{n=1}^{\infty} \frac{2^n + 5}{3^{n+1}}$, $T = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$
 $\downarrow a_n > 0$ $\downarrow b_n > 0$

$$\lim_{n \rightarrow \infty} \frac{2^n + 5}{3^{n+1}} \cdot \frac{3^n}{2^n} = \lim_{n \rightarrow \infty} \frac{1 + \frac{5}{2^n}}{3} = \frac{1 + 0}{3} = \frac{1}{3}$$

$$\therefore \rho \neq 0$$

Since T converges, S also converges by limit comparison test.

3. Ratio Test: Let $S = \sum_{n=M}^{\infty} a_n$, $a_n > 0 \rightarrow$ This is most important

$$\text{let } \rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} > 0$$

- If $\rho < 1$, then S is convergent
- If $\rho > 1$, then S is divergent
- If $\rho = 1$ or doesn't exist, the test is inconclusive.

Example: $S = \sum_{n=1}^{\infty} \frac{2^n + 5}{3^{n+1}}$
 $\downarrow = a_n$

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1} + 5}{3^{n+1}} \cdot \frac{3^n}{2^n + 6}$$

$$= \frac{1 + \frac{5}{2^{n+1}}}{3} \cdot \frac{1}{\frac{1}{2} + \frac{5}{2^{n+1}}}$$

divide top and bottom by 2^{n+1} , powers of 3 cancel.

$$\text{As } n \rightarrow \infty \\ = \frac{1}{3} \cdot 2 = \frac{2}{3} = \rho < 1$$

$\therefore S$ is Convergent by the Ratio Test.

Example 2: $S = \sum_{n=0}^{\infty} \frac{n^2}{2^n}$

$\parallel$
 $a_n > 0$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \frac{1 + \frac{2}{n} + \frac{1}{n^2}}{2}$$

$$\text{As } n \rightarrow \infty \\ = \frac{1}{2} < 1$$

$\therefore S$ Converges by the Ratio Test.

4. nth Root Test (Cauchy)

$$S = \sum_{n=0}^{\infty} a_n \quad a_n > 0$$

$$\text{Let } \rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_n} > 0$$

- If $\rho < 1$, S is Convergent

- If $\rho > 1$, S is Divergent

- If $\rho = 1$ or does not exist, the test is inconclusive.

Note - This test is least important, only works in special cases

Ratio Test fail example:

$$S = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

by Norbert

Try Ratio Test

$$\frac{a_{n+1}}{a_n} = \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \frac{n^2}{n^2 + 2n + 1}$$

$$= \frac{1}{1 + \frac{1}{n} + \frac{1}{n^2}}$$

As $n \rightarrow \infty$

$$= 1$$

$\rightarrow$ Inconclusive. (even tho we know it's convergent by Norbert)

$$S = \sum a_n \quad a_n \geq 0$$

Comparison Test: $T = \sum b_n \quad a_n \leq b_n \quad T \text{ Convergent} \Rightarrow S \text{ Convergent}$

Limit Comparison Test: $T = \sum b_n, b_n > 0$, if $\rho = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ non-zero and finite, then:

$$S \text{ Convergent} \Leftrightarrow T \text{ Convergent}$$

$$S \text{ divergent} \Leftrightarrow T \text{ divergent}$$

Ratio Test: $\rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$, $\rho < 1 \Rightarrow S \text{ Convergent}$
 $\rho > 1 \Rightarrow S \text{ divergent}$

nth root Test: $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_n}$ $\uparrow$

$$S = \sum \frac{1}{n^2} \rightarrow \rho = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$$

$$S = \sum \frac{1}{n} \rightarrow \rho = 1$$

5. Integral Test

$$S = \sum_{n=1}^{\infty} a_n \quad a_n \geq 0$$

If there exists $f: [N, \infty) \rightarrow [0, \infty)$ such that $a_n = f(n)$, with f_k monotone decreasing (i.e. $f(y) \leq f(x)$ for all $y \geq x$), then S is convergent if and only if $\lim_{k \rightarrow \infty} \int_N^k f(x) dx$ exists

Let us apply Integral Test to $S = \sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right) \left(= \frac{\pi^2}{6} \right)$
 $\downarrow a_n \Rightarrow f(x) = \frac{1}{x^2}$

Choose $f: [1, \infty) \rightarrow [0, \infty)$, $f(n) = \frac{1}{n^2} \rightarrow$ monotone decreasing
 $f(n) = a_n$

Integral Test: $\sum$ Converges if and only if $\lim_{K \rightarrow \infty} \int_1^K \frac{dx}{x^2} < \infty$

$$= \lim_{K \rightarrow \infty} \left[-\frac{1}{x} + C \right]_1^K$$

$$= \lim_{K \rightarrow \infty} \left(1 - \frac{1}{K} \right) = 1 < \infty$$

$\therefore$ Convergent

Integral Test for $\sum_{n=1}^{\infty} \frac{1}{n}$ (Divergent)

Choose $f(x) = \frac{1}{x} \geq 0$, monotone decreasing on $[1, \infty)$

Check $\lim_{K \rightarrow \infty} \int_1^K \frac{dx}{x}$

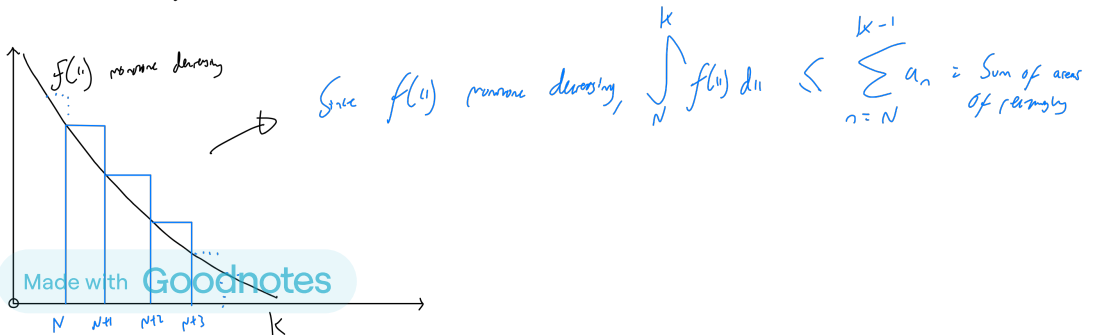
$$= \lim_{K \rightarrow \infty} \left[\log x + C \right]_1^K$$

$$= \lim_{K \rightarrow \infty} (\log K - \log 1)$$

$$= \lim_{K \rightarrow \infty} \log K$$

$$= \infty$$

$\therefore$ Divergent



Convergence Tests for Series with positive and negative Terms

For a series $S = \sum a_n$ with terms with both signs, two notions of convergence:

We say S is absolutely convergent if $\sum |a_n|$ is convergent.

We say S is conditionally convergent if it is convergent but $\sum |a_n|$ diverges.

A series $\sum a_n$ which converges absolutely is also convergent itself:

$$\text{If } \sum_{n=1}^{\infty} |a_n| < \infty \Rightarrow \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n = S_N \text{ exists and is finite.}$$

Note: $\sum |a_n|$ has all $a_n > 0$ so our previous tests can be used (1)

Ratio test for $\sum a_n$ with terms a_n of both signs:

$$\text{Let } \rho = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} \Rightarrow \text{If } \rho < 1 \text{ we know that } \sum_{n=1}^{\infty} |a_n| \text{ is convergent, therefore}$$

$\sum_{n=1}^{\infty} a_n$ is absolutely convergent and therefore convergent.

If $\rho > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent (a_n does not go to zero)

Comparison test:

$$\text{Given } S = \sum a_n \text{ and another } T = \sum b_n$$

If T converges and $|a_n| \leq b_n$ for all $n > N_0$ for some N_0 , then S converges absolutely and is convergent itself.

An alternating series is a series where the signs of the terms alternate, e.g. $S = \sum (-1)^n \cdot |a_n|$

$$\text{Example: } 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \pm \dots$$

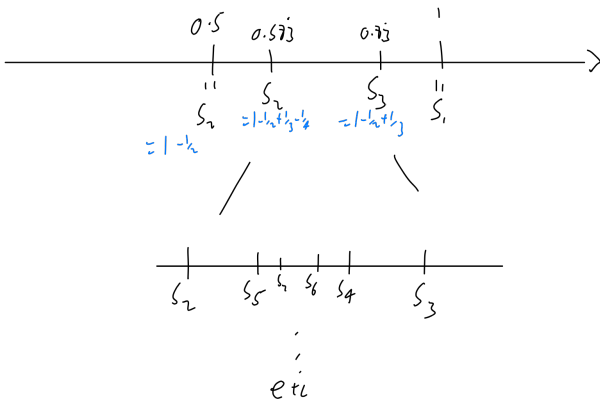
$$= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$$

$$\text{Made with } \text{Goodnotes} \quad \sum_{n=1}^{\infty} \frac{1}{n} \quad |r| < 1$$

$$\Rightarrow |1| < 1 : \ln 1 = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \quad \text{If we could choose } |1| = 1 \Rightarrow \ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

dodgy 'alt math' $\therefore \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = -\ln 2 = \ln\left(\frac{1}{2}\right)$

$$S = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}, \quad S_N = \sum_{k=1}^N \frac{(-1)^{k+1}}{k}$$



$$S_1 = 1$$

$$S_2 = 0.5$$

$$S_3 = 0.833$$

$$S_4 = 0.583$$

⋮

$$S_{1000} = 0.6926474$$

$$S_{1001} = 0.6936464$$

$$S_{1002} = 0.6926472$$

⋮

⋮

Converges to $\ln 2 = 0.693147$

S not absolutely convergent.

$$\sum_{k=1}^{\infty} \frac{1}{k} = \infty \rightarrow S \text{ only conditionally convergent.}$$

Alternating Sign Test.

$$S = \sum_{n=1}^{\infty} (-1)^n a_n \quad a_n \geq 0$$

a_k does not include the $(-1)^n$ bit

S converges, at least conditionally, if a_k monotonically decreasing, i.e. $a_{k+1} \leq a_k$ for all k and $a_k \rightarrow 0$

Example: $S = \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$

Made with Goodnotes
 $\frac{1}{(k+1)^2} \leq \frac{1}{k^2} \rightarrow$ monotonically decreasing ✓

$$\frac{1}{k^2} \rightarrow 0 \text{ as } k \rightarrow \infty \checkmark$$

Alternating Sign test $\Rightarrow S$ is Convergent

We also have $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ So we have absolute convergence of S .

$\sum a_k$ is absolutely convergent if $\sum |a_k|$ is convergent
 $\Rightarrow \sum a_k$ is convergent.

Power Series and Taylor Series

Already considered $\sum_{n=0}^{\infty} a_n(x)$.

Special case: $\sum_{n=0}^{\infty} \underbrace{b_n x^n}_{a_n(x)}$ is called a Power Series.

We can think of power series as representing functions in $x \in \mathbb{R}$

We know $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$ for all $|x| < 1 \rightarrow$ this is a power series

Then $S_3 = 1 + x + x^2$ is an approximation of $\frac{1}{1-x}$, as are S_{10}, S_{100} etc.

We know $\ln(1-x) = - \sum_{k=1}^{\infty} \frac{1}{k} x^k$ is also a power series

Convergence of a Power Series:

$$P(x) = \sum_{n=0}^{\infty} \underbrace{b_n x^n}_{a_n(x)}$$

$$\text{Ratio Test: } \lim_{n \rightarrow \infty} \frac{a_{n+1}(x)}{a_n(x)} = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1} x^{n+1}}{b_n x^n} \right|$$

$$= |x| \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right|$$

$$\text{Let } \rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right|$$

$$\text{If } |x| > \frac{1}{\rho} \Rightarrow P(x) \text{ divergent}$$

$$|x| < \frac{1}{\rho} \Rightarrow P(x) \text{ convergent}$$

Let $P(x) = \sum_{n=0}^{\infty} b_n x^n$ be a power series

Made with Goodnotes

$$\text{Let } \rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right|$$

Then $P(x)$ diverges for $|x| < \frac{1}{\rho}$
 $P(x)$ converges for $|x| > \frac{1}{\rho}$

gang ses

If $\rho = 0 \rightarrow$ convergent

$\rho = \infty \rightarrow$ divergent

We call $\frac{1}{\rho}$ the radius of convergence of the power series

Let $P(x) = \sum_{n=0}^{\infty} b_n x^n$ be a power series. Then there exists $\rho \in [0, \infty]$ Inclusive

such that $P(x)$ converges for all $|x| < \frac{1}{\rho}$ and diverges for all $|x| > \frac{1}{\rho}$

Convergence / Divergence at $x = -\frac{1}{\rho}$ and $x = \frac{1}{\rho}$ need to be investigated separately.

All $x \in \mathbb{R}$ for which $P(x)$ converges form the interval of convergence of $P(x)$

Example: $P(x) = \sum_{n=0}^{\infty} x^n$

$b_n x_n = x^n$
 $\therefore b_n = 1$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = 1$$

$\Rightarrow$ radius of convergence $\frac{1}{\rho} = 1$

for $x = 1$, $P(x) = \sum_{n=0}^{\infty} 1^n = \infty$

for $x = -1$, $P(x) = \sum_{n=0}^{\infty} (-1)^n$ not convergent

$\therefore$ Interval of Convergence = $(-1, 1)$

non-inclusive interval.

Example: $P(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$

$\hookrightarrow$ this power series represents e^x

$$b_n = \frac{1}{n!} \Rightarrow \rho = \lim_{n \rightarrow \infty} \left| \frac{b_n}{b_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

$$\Rightarrow \text{radius of convergence } \frac{1}{\rho} = \infty$$

$\Rightarrow P(x)$ converges for all $x \in \mathbb{R}$

Example: $P(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$

$$b_0 = 0, b_n = \frac{1}{n}, n \geq 1$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| = 1$$

$$\Rightarrow \text{Radius of convergence of } P(x), \frac{1}{\rho} = 1$$

For $x = -1$: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \rightarrow \text{Convergent (Alternating Series)}$

for $x = 1$: $\sum_{n=1}^{\infty} \frac{1}{n} \rightarrow \text{Divergent (Harmonic Series)}$

$$\Rightarrow \text{Interval of Convergence of } P(x) \text{ is } [-1, 1)$$

$$\hookrightarrow -1 \leq x < 1, x \in \mathbb{R}$$

Example: $P(x) = \sum_{n=0}^{\infty} \frac{x^n}{n^2}$

$$e = 1$$

$$\text{Interval of Convergence } [-1, 1]$$

General Power Series $S(x) = \sum_{n=0}^{\infty} b_n (x-a)^n \quad a \in \mathbb{R}$

For any general power series, there exists an $r \in [0, \infty]$ such that $S(x)$ converges absolutely for

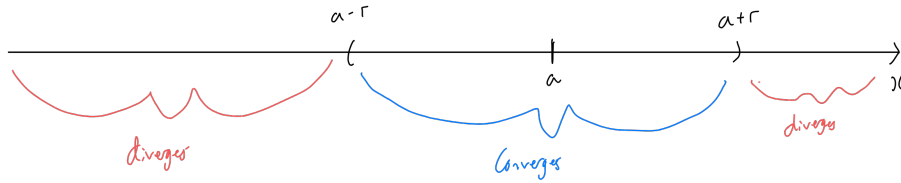
all $|x-a| < r$ and $S(x)$ diverges for all $|x-a| > r$

$r =$ Radius of Convergence

Very often, we can compute r as follows:

If $\rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right|$ exists and is in $[0, \infty]$, then:

$$r = \frac{1}{\rho} \quad \text{with } r=0 \text{ if } \rho = \infty \\ r = \infty \text{ if } \rho = 0$$



We can differentiate and integrate power series termwise and we obtain the derivative and integral of $S(x)$ and the new power series has the same radius of convergence.

If $S(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$ has radius of convergence $r = \frac{1}{\rho}$ with $\rho = \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right|$, we have

$$S'(x) = \sum_{n=1}^{\infty} b_n n (x-a)^{n-1} = \sum_{m=0}^{\infty} b_{m+1} (m+1) (x-a)^m$$

like differentiating, constant term b_0 needs to be deleted so go from 1 to ∞

$$\Rightarrow \text{radius of convergence of } S'(x) = \lim_{m \rightarrow \infty} \frac{b_{m+1} (m+1)}{b_{m+1} (m+1)} = \lim_{m \rightarrow \infty} \left| \frac{b_{m+1}}{b_{m+1}} \right| \times \lim_{m \rightarrow \infty} \left| \frac{m+1}{m+1} \right|$$

this tends to 1

$$= \lim_{m \rightarrow \infty} \left| \frac{b_{m+1}}{b_{m+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \rho$$

$\therefore$ radius of convergence of $S'(x) = r' = \frac{1}{\rho} = r$

$\therefore$ Same radius of convergence for termwise differentiation and integration.

Example: $\frac{1}{1-u} = \sum_{k=0}^{\infty} u^k$

replace u with $-u$
 $\rightarrow$ absolutely convergent for $|u| < 1$
 divergent for $|u| > 1$
 $r = 1$

$$\frac{1}{1-(-u)} = \sum_{k=0}^{\infty} (-u)^k = \sum_{k=0}^{\infty} (-1)^k u^k$$

replace u with y
 $\rightarrow$ convergent for $|u| < 1$
 divergent for $|u| > 1$
 $r = 1$

$$\frac{1}{1+y^2} = \sum_{k=0}^{\infty} (-1)^k (y^2)^k = \sum_{k=0}^{\infty} (-1)^k y^{2k}$$

$\rightarrow$ convergent for $|y| < 1 \rightarrow$ also for $|y| < 1$
 divergent for $|y| > 1 \rightarrow$ also for $|y| > 1$

replace y with $2z$

$$\frac{1}{1+4z^2} = \sum_{k=0}^{\infty} (-1)^k (2z)^{2k} = \sum_{k=0}^{\infty} (-1)^k 4^k z^{2k} = \sum_{k=0}^{\infty} (-4)^k z^{2k}$$

$$|y| = |2z| < 1 \text{ convergent} \rightarrow |z| < \frac{1}{2}$$

$$|y| = |2z| > 1 \text{ divergent} \rightarrow |z| > \frac{1}{2}$$

$$r = \frac{1}{2}$$

Differentiate:

$$\begin{aligned} \frac{8z}{(1+4z^2)^2} &= \sum_{k=1}^{\infty} (-4)^k 2k z^{2k-1} \\ &= \sum_{L=0}^{\infty} (-4)^L 2(L+1) z^{2L+1} \end{aligned}$$

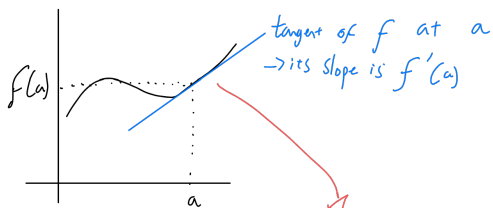
$r = \frac{1}{2}$ since r does not change when differentiating termwise.

Taylor Series

Given an infinitely many times differentiable function f :

Aim - Find a power series $\sum_{n=0}^{\infty} b_n(x-a)^n$ representing f near a . (Taylor Series)

Best linear approximation of a function f , differentiable at a , near a .



Tangent is a linear function: $y = f(a) + (x-a)f'(a)$

This is the best linear approximation of f near a .

This is also the first Taylor polynomial of f at a :

$$p_1(x) = p_{1,a}(x) = f(a) + (x-a)f'(a)$$

the error is $f(x) - p_1(x) = f(x) - f(a) - f'(a)(x-a)$

$$= x-a \left[\frac{f(x)-f(a)}{x-a} - f'(a) \right] \rightarrow \text{as } x \rightarrow a, \underbrace{\frac{f(x)-f(a)}{x-a}}_{\text{this } \rightarrow f'(a) \text{ as } x \rightarrow a} - f'(a) \rightarrow 0$$

$$\Rightarrow \frac{f(x) - p_1(x)}{x-a} \rightarrow 0 \text{ as } x \rightarrow a \text{ where } p_1 \text{ is a linear approximation of } f \text{ at } a.$$

n th Taylor polynomial is a polynomial of degree n satisfying $\frac{f(x) - p_n(x)}{(x-a)^n} \rightarrow 0 \text{ as } x \rightarrow a$

Computation of second Taylor polynomial:

$$\begin{aligned} f(x) &= f(a) + \int_a^x f'(y) dy \approx f(a) + \int_a^x f'(a) + (y-a)f''(a) dy \\ &\quad \text{replace } f'(y) \text{ with its} \quad = f(a) + \underbrace{f'(a)(x-a)}_{\text{extension of } \int_a^x f'(a) dy} + f''(a) \int_a^x (y-a) dy \\ &\quad \text{(i.e. Taylor polynomial)} \end{aligned}$$

$$p_2(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$$

For f infinitely many times differentiable at a :

N th Taylor polynomial of f at a is the "best" polynomial approximation of degree N of f near a .

$$p_N(x) = p_{N,a}(x) = \sum_{n=0}^N \frac{f^{(n)}(a)}{n!} (x-a)^n$$

Properties:

- We have $\lim_{x \rightarrow a} \frac{f(x) - p_N(x)}{(x-a)^N} = 0$ ← top goes to 0 faster than $(x-a)^N$ as $x \rightarrow a$

- We also write $f(x) - p_N(x) = o((x-a)^N)$ as $x \rightarrow a$ Landau o -notation (little- o) Why is o -notation in maths ???

In general $h(x) = o(g(x))$ as $x \rightarrow a$ means $\lim_{x \rightarrow a} \frac{h(x)}{g(x)} = 0$

- $p_N^{(k)}(a) = f^{(k)}(a)$ $k \leq N$
(if $k > N$, $p_N^{(k)} = 0$ since p_N is an N th degree polynomial)

Examples

(1) $f(x) = \cos x$, $a = 0$

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f'''(x) = \sin x$$

$$f^{(4)}(x) = \cos x = f(x) \rightarrow \begin{array}{ll} f^{(4k)}(x) = \cos x & f^{(4k)}(0) = 1 \\ f^{(4k+1)}(x) = -\sin x & f^{(4k+1)}(0) = 0 \\ f^{(4k+2)}(x) = -\cos x & f^{(4k+2)}(0) = -1 \\ f^{(4k+3)}(x) = \sin x & f^{(4k+3)}(0) = 0 \end{array}$$

$$p_4(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2} + \frac{f'''(0)x^3}{6} + \frac{f^{(4)}(0)x^4}{24}$$

$$= 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$$

= best polynomial approximation of $\cos x$ of degree 4 near 0.

$$p_N(x) = \sum_{k=0}^{N/2} \frac{f^{(2k)}(0)}{(2k)!} x^{2k} \quad 2k \leq N$$

$$(2) f(x) = \ln x, \quad a = 1$$

$$f'(1) = \frac{1}{1}$$

$$f''(1) = -\frac{1}{1^2}$$

$$f'''(1) = \frac{2}{1^3}$$

$$f^{(4)}(1) = -\frac{6}{1^4}$$

$$f^{(n)}(1) = (-1)^{n+1} \cdot \frac{(n-1)!}{1^n} \quad n \geq 1$$

$$P_{N,1}(x) = \sum_{n=1}^N \frac{(-1)^{n+1} (n-1)!}{n!} (x-1)^n = \sum_{n=1}^N \frac{(-1)^{n+1}}{n} (x-1)^n$$

If f is itself a polynomial of degree n , that is:

$$f(x) = a_0 + a_1(x-a) + a_2(x-a)^2 + \dots + a_n(x-a)^n$$

then

$$P_{N,a}(x) = f(x) \text{ for } N \geq n, \quad P_{N,a}(x) = a_0 + a_1(x-a) + \dots + a_N(x-a)^N \text{ for } N < n$$

Example:

$$f(x) = x^3 + 5x + 1, \quad a = 0$$

$$f'(x) = 3x^2 + 5$$

$$f''(x) = 6x$$

$$f'''(x) = 6$$

$$f^{(4)}(x) = 0$$

$$f(0) = 1$$

$$f'(0) = 5$$

$$f''(0) = 0$$

$$f'''(0) = 6$$

$$f^{(4)}(0) = 0$$

$$P_1(x) = f(0) + f'(0)x$$

$$= 1 + 5x$$

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2$$

$$= 1 + 5x$$

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{6}x^3$$

$$= 1 + 5x + x^3$$

$$= f(x)$$

The Taylor Series of a function $f(x)$ about $x = a$ is a power series:

$$P_a(x) = \lim_{N \rightarrow \infty} P_{N,a}(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

There exists a radius of convergence $r \in [0, \infty]$

$P_n(x)$ has an interval of convergence and, in most cases, we have $f(x) = P_n(x)$ in the interval of convergence (but not always).
 $\rightarrow$ all cases we need to worry about $\odot$

If this is true, we say "f is analytic in a". All functions will be analytic in this course $\odot$

If $a=0$, Taylor Series are also sometimes called Maclaurin Series

Function	Taylor Series
e^x	$\sum_{k=0}^{\infty} \frac{x^k}{k!}$
$\cos x$	$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \rightarrow$ only even powers of x $\Rightarrow \cos x$ is an even function $\rightarrow \cos(-x) = \cos x$
$\sin x$	$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \rightarrow$ only odd powers of x $\Rightarrow \sin x$ is an odd function $\rightarrow \sin(-x) = -\sin x$
$\ln(1+x)$	$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k$

} Interval of convergence = $\mathbb{R}$

) Interval of convergence $[-1, 1]$

Nth Taylor polynomial of f at a : $P_N(x) = P_{N,a}(x) = \sum_{k=0}^N \frac{f^{(k)}(a)}{k!} (x-a)^k$

Taylor Series of f at a : $P(x) = P_a(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$

$\rightarrow$ is a power series with interval of convergence

In most cases: $P_n(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$ agrees with $f(x)$ within the interval of convergence:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k \text{ for } |x-a| < r \text{ where } r = \text{radius of convergence.}$$

Taylor's Theorem: Error estimate for $\left| f(x) - \sum_{k=0}^N \frac{f^{(k)}(a)}{k!} (x-a)^k \right|$

f $(N+1)$ -times differentiable near a and x a nearby point to a . Assume $f^{(N+1)}$ continuous between a and x .

Let $P_{N,a}(x) = \sum_{k=0}^N \frac{f^{(k)}(a)}{k!} (x-a)^k$ and

$R_{N,a}(x) = f(x) - P_{N,a}(x)$ the error.

there exists ξ between x and a such that:

$$R_{N+1,a}(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x-a)^{N+1}$$

$R_{N+1}(x)$ is called the N^{th} remainder term of f near a .

The RHS is called the Lagrange form of the remainder

If f itself is a polynomial of degree N , then $f(x) = p_N(x)$ for all $m > N$
 $\hookrightarrow f^{(N+1)}(\xi) = 0$ for all ξ

$\therefore R_{N+1,a}(x) = 0$ in this case.

The crucial point is that we don't know ξ between x and a , but if we know $|f^{(N+1)}(y)| \leq M$ for all y between x and a , then $|f(x) - p_{N,a}(x)| \leq \frac{M}{(N+1)!} (x-a)^{N+1}$ providing an error estimate.

Example: $f(x) = e^{\cos x}$, find $p_2(x)$ about $a = \frac{\pi}{2}$ and get an error bound

$$f(x) = e^{\cos x}$$

$$f'(x) = -\sin x e^{\cos x}$$

$$f''(x) = (-\cos x + \sin^2 x) e^{\cos x}$$

$$f'''(x) = (\sin x + 3\sin x \cos x - \sin^3 x) e^{\cos x}$$

$$f\left(\frac{\pi}{2}\right) = 1$$

$$f'\left(\frac{\pi}{2}\right) = -1$$

$$f''\left(\frac{\pi}{2}\right) = 1$$

$$p_{2,\pi/2}(x) = f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)(x - \frac{\pi}{2}) + \frac{f''\left(\frac{\pi}{2}\right)}{2}(x - \frac{\pi}{2})^2$$

$$= 1 - (x - \frac{\pi}{2}) + \frac{1}{2}(x - \frac{\pi}{2})^2$$

$$= 1 + \frac{\pi}{2} - x + \frac{1}{2}(x - \frac{\pi}{2})^2$$

$$R_{N+1,\pi/2}(x) = \frac{f^{(3)}(\xi)}{6} (x-a)^3 = \frac{1}{6} (\underbrace{\sin \xi}_1 + \underbrace{3\sin \xi \cos \xi}_3 - \underbrace{\sin^3 \xi}_1) e^{\cos \xi} (x-a)^3 \text{ for some } \xi \text{ between } x \text{ and } a = \frac{\pi}{2}$$

$$e^{\cos x} - p_{2,\pi/2}(x)$$

$$\text{Approx. error} \quad |R_{N+1,\pi/2}(x)| \leq \frac{1}{6} \times 5 \times e (x - \frac{\pi}{2})^3 = \frac{5e}{6} (x - \frac{\pi}{2})^3$$

$$\sin \xi + 3\sin \xi \cos \xi - \sin^3 \xi = \sin \xi (1 - \sin^2 \xi) + 3\sin \xi \cos \xi$$

$$= \sin \xi \cos^2 \xi + 3\sin \xi \cos \xi$$

$$= S_{1,1} \{ \cos \} (\cos \{ + 3 \})$$

$$= \frac{S_{1,1}(25)}{2} (\cos \{ + 3 \})$$

$$\downarrow$$

max abs. value

$$= \frac{1}{2} \times 4 = 2$$

$$|R_{N_H, \pi_2}(u)| \leq \frac{1}{6} \times e (u - \pi_2)^3 = \frac{e}{3} (u - \pi_2)^3$$

$$\pi_2 \approx 1.5708 \dots$$

$$u = 1.5 \rightarrow |e^{u_0} - p_{2, \pi_2}(u)| \leq \frac{e}{3} |u - \pi_2|^3 \approx \underline{0.000322}$$

error estimate

$$p_{2, \pi_2}(1.5) = 1 + \pi_2 - 1.5 + \frac{1}{2} (1.5 - \pi_2)^2 \approx 1.073302$$

$$e^{\cos(1.5)} \approx 1.073299$$

$$\text{actual error } |1.073299 - 1.073302|$$

$$= 0.000003$$